

WYVERN-E 4.0 — Research Proposal

A Skylight Rocketry Venture · 70 mm Single-Stage Servo-TVC Sustainer

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Executive Summary

WYVERN-E 4.0 is a 70 mm, single-stage, **3D-printed active-thrust-vector-control (TVC)** research rocket that demonstrates closed-loop flight stabilization on a bare-metal **Teensy 4.1** flight computer, powered by the **Estes F15-0**. It carries four small fins plus 58 mm fins (no ballast) for **passive stability during launch and the F15 ignition spike**, after which ($t \geq 0.5$ s) the TVC loop engages on the smooth portion of the thrust curve to stabilize and execute a commanded maneuver. The two candidate TVC actuators — a tri-solenoid *magnetic* gimbal and a *servo* gimbal — are compared quantitatively on a purpose-built **3-axis thrust-vector balance** that resolves both thrust magnitude and vector direction; the flight vehicle carries the servo system. The program retains the WYVERN aerofoil wind-tunnel (RQ1/RQ2) and static motor/materials test regimens. All flights are FAA Class-1 (no waiver, no certification). Predicted apogee ≈ 432 ft; project cost $\approx \$2,040$.

1. Background & Motivation

1.1 Thrust vector control

Thrust vectoring steers a rocket by gimbaling the exhaust, producing a control moment without aerodynamic surfaces — the method used by every orbital launch vehicle. A small, low-cost, fully printed TVC vehicle makes the control problem tangible while producing real flight data.

1.2 WYVERN lineage

WYVERN-E 1.0 (interceptor study) → 2.0 (84 mm two-stage, custom avionics) → 3.0 (Raspberry Pi 5, solenoid-vs-servo A/B in flight) established the aero, structures, and avionics groundwork. 4.0 distills those lessons into the simplest vehicle that still answers the core question: a single stage, a single bare-metal controller, and the A/B actuator comparison moved to a repeatable ground balance.

1.3 The 4.0 thesis

A small finned rocket can use passive stability to survive launch and the ignition transient, then hand authority to a closed-loop TVC system on the smooth thrust curve — and the relative merit of magnetic vs servo gimbals can be measured directly on the ground before flight.

2. Research Questions

#	Question	Method	Primary metric
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RQ1	Aerofoil lift/drag & deflection behaviour	Hofferth wind tunnel, 0.5° sweeps	Cl, Cd, Cl/Cd vs α
RQ2	Print-material flame/erosion as jetvane candidates	F15-0 plume on static stand + deflector	mass-loss, char depth
RQ3	Magnetic vs servo TVC (ground)	3-axis thrust-vector balance	bandwidth, slew, overshoot, SSE, max vector angle
RQ4	Closed-loop flight stabilization + maneuver	onboard log, 3 flights	pitch error, gimbal track, recovery

3. Vehicle Architecture

3.1 Configuration & mass budget

70 mm OD, 0.74 m, three bays + two bulkheads; 4 fins + nose ballast.

Section	Key mass	Subtotal
Nose (PC-FR)	cone 40 g	40 g
Recovery bay (ASA-Aero)	RRC3+, 9 V, chute, e-match, BNO085	118 g
FC bay (ASA-Aero)	Teensy 4.1, BNO085, BME688, BMP280, μ SD, RunCam, 2S LiPo, BEC	55 g
Engine/TVC bay (PC-FR)	gimbal, 2 servos, BNO085, mount	152 g
Structure	body tube + 4 PC-FR fins + wiring	203 g
Dry		606 g
Motor	F15-0 (60 g prop)	102 g
Liftoff		708 g

3.2 Materials

- **PC-FR** (ρ 1.25 g/cm³, HDT ~ 110 °C, UL94-V0): nose, engine/TVC bay, Bulkhead A, fins — flame zone.
- **ASA-Aero** (foamed, ρ ~0.65 g/cm³): body, FC/recovery bays, Bulkhead B — saves ~100 g where there is no motor heat. Thermal check: engine-bay wall peaks **~38 °C** over the 3.45 s burn (no liner needed; first-order lumped model, we4_analysis.py).

3.3 Stability — fins + ballast + the 0.5 s rule

An apogee sweep shows ballast lowers altitude, so we use **no ballast** and size fins to the minimum stable 1.0 cal: 4 × 58 mm fins — CP 52.5 cm, CG 46.7 cm = **+1.0 cal** static margin (stable). This passive margin holds the vehicle through launch and the F15 ignition spike; the TVC controller is **inhibited until t = 0.5 s**, then engages on the smooth curve. A finless variant (margin −5.6 cal) was rejected because it is statically unstable and cannot survive the pre-TVC transient.

3.4 Structural & thermal margins

First-order analysis (we4_analysis.py): minimum safety factor **> 300×** (the 25 N motor leaves the 1.6 mm airframe print/handling-limited, not load-limited); fin flutter velocity well above the 25 m/s flight regime; engine-bay thermal margin to PC-FR HDT > 70 °C.

4. Propulsion & Trajectory

4.1 Motors (verified)

Motor	Spec	Role	Qty
Estes F15-0	49.6 N·s, 14.4 N avg / 25.3 N pk, 3.45 s, 0-delay	flight + both stands + jetvane	16
Estes E16-4	~16 N avg	stand commissioning	6

4.2 Predicted performance (unified RK4 + Barrowman, we4_flightsim.py)

T/W 2.07 avg / 3.62 peak; Cd 0.53; burnout 3.45 s, ~73 m, ~30 m/s; **apogee ~432 ft @ 7.0 s**; deploy forced t = 4.0 s @ ~29 m/s; 18" chute → ~6 m/s descent. Dispersion (±5 % mass, ±15 % Cd): see plots4/06_dispersion.png.

5. Flight Computer & Control

5.1 Teensy 4.1

A single 600 MHz Cortex-M7 is flight computer *and* 500 Hz controller: reads three BNO085 + two baros, closes the TVC loop, drives 2 servos via hardware FlexPWM, logs to onboard SDIO microSD, and triggers recovery/camera. No Linux, no scheduler jitter.

5.2 IMUs — Game Rotation Vector

All three BNO085 run in **Game Rotation Vector** (accel + gyro, magnetometer disabled) to reject the magnetic interference of the adjacent servos. Gimbal deflection = $q_{body}^{-1} \otimes q_{gimbal}$, giving true nozzle attitude relative to the body (catches linkage backlash/flex). The recovery-bay unit votes against the FC unit for fault detection.

5.3 Control law

Per-axis PID ($K_p=8$, $K_i=1.5$, $K_d=1.2$), output clamped to $\pm 5^\circ$, servo lag $\tau \approx 0.04$ s. **TVC inhibited for the first 0.5 s** (ignition spike) — fins hold attitude — then engages: stabilize to vertical, then a 4° commanded maneuver. Required gimbal torque 0.56 kg·cm (micro-servo class); control authority positive throughout the powered phase.

5.4 Power & data

2S 450 mAh LiPo — 5 V BEC (Teensy/IMUs/camera) + servo rail; independent 9 V lithium for the RRC3+. Active draw 5.7 W — ~ 32 min endurance. Onboard log: full-rate IMU/baro/control + RunCam 1080p video.

6. Recovery

Finless-after-burnout dynamics demand early deploy: RRC3+ **Aux timer at $t = 4.0$ s** (Event=Launch $\rightarrow$ Timer $\rightarrow$ 1-shot), arming altitude 25–50 ft so it latches at liftoff; **drogue-apogee channel as a redundant backup** (second e-match, same charge well, one E-Match Mate canister/launch). Opening at ~ 19 m/s; 1/8" Kevlar cord ($> 60\times$ margin) with elastic leader; 18" chute $\rightarrow \sim 6$ m/s.

7. Ground Test Program

7.1 3-axis thrust-vector balance

Motor + gimbal on a thrust block restrained by one axial (5 kg) + two lateral (1 kg) strain-gauge load cells — T, θ, ϕ . Actuator-agnostic — both magnetic and servo systems tested identically. RQ3 metrics logged vs commanded. ### 7.2 Static thrust + materials stand Axial cell + steel deflector: validates the F15-0 curve and screens jetvane materials in the plume. ### 7.3 Motor & calibration plan Load cells dead-weight calibrated, then commissioned with **6 $\times$ E16-4** (3 per stand). Counts: F15-0 $\times 16$, E16-4 $\times 6$.

8. Wind Tunnel (RQ1/RQ2)

Hofferth modular STEM tunnel with the AC Infinity Cloudline A8 (724 CFM) for force-capable test-section velocity; fin articles on the Gridfinity strut/sting and sidewall half-span mounts.

9. Safety & Regulatory

Single F15-0: 49.6 N·s, 60 g propellant, $\leq$ F class, liftoff 708 g < 1500 g — **FAA Class-1, no waiver, no Level-1 certification**. Remote ignition, ≥ 3 m standoff on the stands, gimbal-neutral fail-safe, independent recovery-bay arming, RBF pyro inhibit.

10. Budget

$\approx$ **\$2,040** total (vehicle + 3-axis balance + static/materials stand + wind tunnel + one-time tools + all motors); 65-line BOM with live links in

Documentation/WYVERN_E4_BOM.xlsx. Per-flight consumable $\approx$ F15-0 \$17 + initiator + BP $\approx$ \$25.

11. Schedule & Milestones

Phase

Print + assemble

Bench bring-up

TVC balance A/B (RQ3)

Static fires + jetvane (RQ2)

Wind tunnel (RQ1)

Flight tests (RQ4)

Analysis + paper

12. Risk Register

Risk	Likelihood	Mitigation
Pre-TVC instability	Med	fins + ballast (+1.5 cal); TVC inhibit 0.5 s
Servo slew too slow	Med	bench-verify on balance before flight; fast digital micro
Hard 19 m/s deploy	Low	Kevlar > 60 $\times$ margin, elastic leader, ground-tested charge
Launch-detect miss at low T/W	Low	tune arming alt; verify on 2.2 g spike
Camera/SD throughput	Low	self-contained RunCam (decoupled from FC)

13. Expected Outcomes & Deliverables

A flight-validated small TVC vehicle; a quantitative magnetic-vs-servo TVC dataset; aerofoil polars and jetvane material rankings; full open documentation (CAD, firmware, wiring, sims) and a research paper.

References

Hofferth, J. *Modular Wind Tunnel for STEM Education*, AIAA SCITECH 2025, doi:10.2514/6.2025-2560. Barrowman, J. *The Practical Calculation of the Aerodynamic Characteristics of Slender Finned Vehicles*, NASA, 1967. Box, J. (BPS.space) *Signal/Echo* TVC flight series. NAR/Tripoli Model Rocket Safety Codes. Bosch Sensortec BNO085 datasheet; PJRC Teensy 4.1 reference.